HDOC Day-42 Assignment

Question: Write a menu-driven C program to perform the following operations using separate functions:

- Remove all vowels from a string.
- Find the first repeating lowercase alphabet in a string.
- Check if two strings are anagrams of each other.

1. Problem Understanding

The program is menu-driven and performs three string operations using separate functions. The user can remove all vowels from a string, find the first repeating lowercase alphabet, or check whether two strings are anagrams. The menu continues to appear until the user selects Exit.

The program uses simple arrays, loops, character checking, and separate functions so that the logic remains clear and suitable for a beginner-level C program.

2. Standard Test Cases

S.No.	Operation / All Inputs	Expected Output
1	Choice = 1 String: Education	String after removing vowels: dctn
2	Choice = 2 String: programming	First repeating lowercase alphabet: r
3	Choice = 3 String 1: listen String 2: silent	The strings are anagrams.
4	Choice = 2 String: abcde	No repeating lowercase alphabet found.
5	Choice = 3 String 1: hello String 2: world	The strings are not anagrams.

3. Algorithm

1. Start.

2. Display the menu containing the three string operations and Exit.

3. Read the user's choice.

4. For choice 1, read a string and call removeVowels(). Copy only non-vowel characters to a result string and display it.

5. For choice 2, read a string and call firstRepeatingLowercase(). Count lowercase letters and display the first character whose count becomes 2.

6. For choice 3, read two strings and call checkAnagram(). Compare their lengths and character frequencies.
7. For choice 4, end the program.
8. For any other choice, display an invalid-choice message.
9. Repeat the menu until Exit is selected.
10. Stop.

4. C Program

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>
void removeVowels(char str[])
{
  int i, j = 0;
  char result[200];

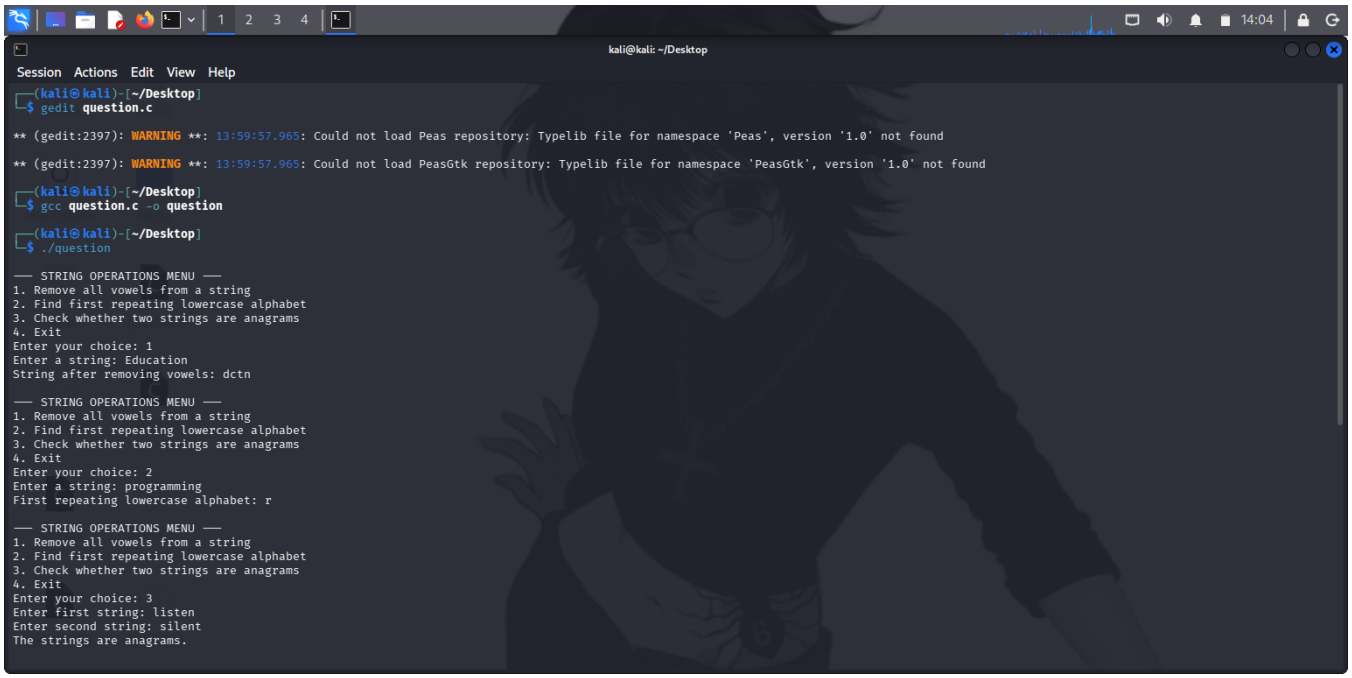
  for (i = 0; str[i] != '\0'; i++)
  {
    char ch = tolower((unsigned char)str[i]);
    if (ch != 'a' && ch != 'e' && ch != 'i' && ch != 'o' && ch != 'u')
    {
      result[j] = str[i];
      j++;
    }
  }
  result[j] = '\0';
  printf("String after removing vowels: %s\n", result);
}
void firstRepeatingLowercase(char str[])
{
  int count[26] = {0};
  int i;
  for (i = 0; str[i] != '\0'; i++)
  {
    if (str[i] >= 'a' && str[i] <= 'z')
    {
```

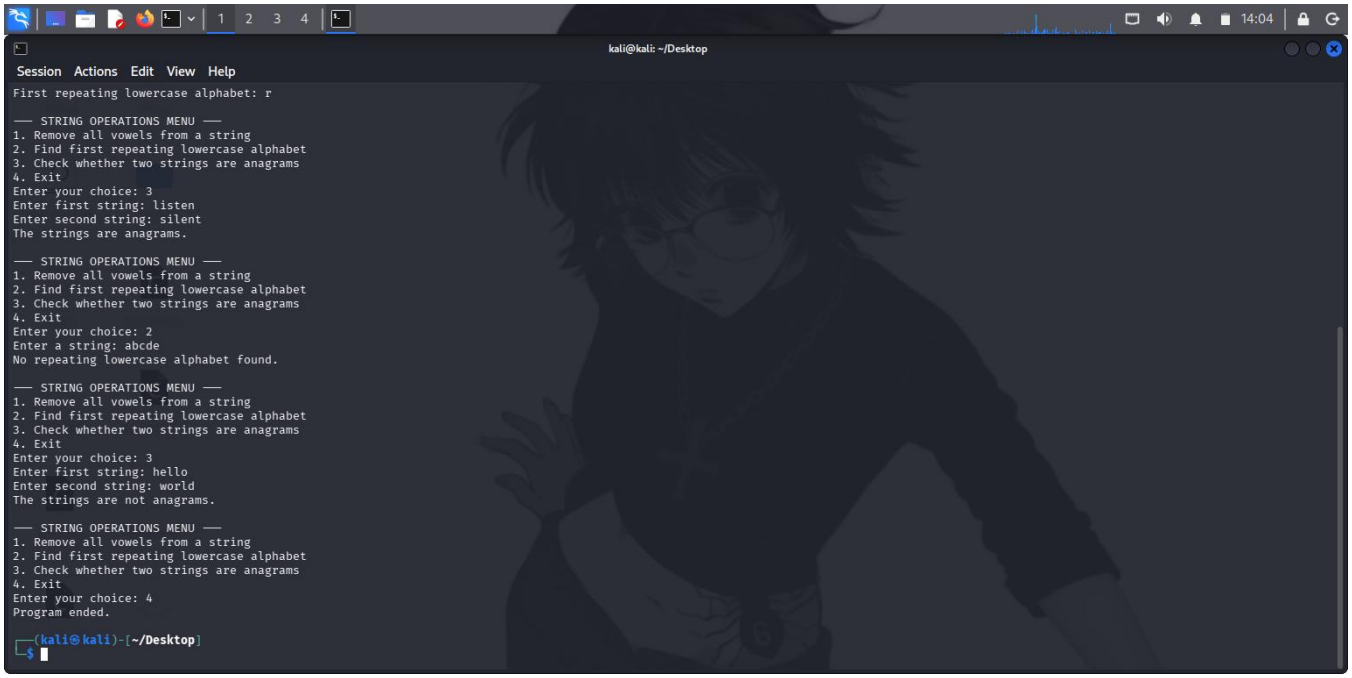
```
            count[str[i] - 'a']++;
            if (count[str[i] - 'a'] == 2)
            {
                printf("First repeating lowercase alphabet: %c\n", str[i]);
                return;
            }
        }
    }
    printf("No repeating lowercase alphabet found.\n");
}
void checkAnagram(char str1[], char str2[])
{
    int count[256] = {0};
    int i;
    if (strlen(str1) != strlen(str2))
    {
        printf("The strings are not anagrams.\n");
        return;
    }
    for (i = 0; str1[i] != '\0'; i++)
    {
        count[(unsigned char)str1[i]]++;
        count[(unsigned char)str2[i]]--;
    }
    for (i = 0; i < 256; i++)
    {
        if (count[i] != 0)
        {
            printf("The strings are not anagrams.\n");
            return;
        }
    }
    printf("The strings are anagrams.\n");
}
```

```
int main()
{
    int choice;
    char str1[200], str2[200];
    do
    {
        printf("\n--- STRING OPERATIONS MENU ---\n");
        printf("1. Remove all vowels from a string\n");
        printf("2. Find first repeating lowercase alphabet\n");
        printf("3. Check whether two strings are anagrams\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        getchar();
        switch (choice)
        {
            case 1:
                printf("Enter a string: ");
                fgets(str1, sizeof(str1), stdin);
                str1[strcspn(str1, "\n")] = '\0';
                removeVowels(str1);
                break;
            case 2:
                printf("Enter a string: ");
                fgets(str1, sizeof(str1), stdin);
                str1[strcspn(str1, "\n")] = '\0';
                firstRepeatingLowercase(str1);
                break;
            case 3:
                printf("Enter first string: ");
                fgets(str1, sizeof(str1), stdin);
                str1[strcspn(str1, "\n")] = '\0';
                printf("Enter second string: ");
                fgets(str2, sizeof(str2), stdin);
```

```
            str2[strcspn(str2, "\n")] = '\0';
            checkAnagram(str1, str2);
            break;
        case 4:
            printf("Program ended.\n");
            break;
        default:
            printf("Invalid choice. Please try again.\n");
    }
  }
  while (choice != 4);
  return 0;
}
```

5. Compilation and Execution Evidence





6. Verification of All Test Cases

Test	Input Summary	Actual Output	Result
1	Education	dctn	Pass
2	programming	First repeating lowercase: r	Pass
3	listen, silent	Strings are anagrams	Pass
4	abcde	No repeating lowercase alphabet	Pass
5	hello, world	Strings are not anagrams	Pass